Bahria University, Islamabad

Department of Software Engineering

Web Engineering Lab (Spring-2026)

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Enrollment : 01-131232-089

Lab Journal: 07

Date: 12th March, 2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Lab No: 07 - React Props, State, and Callback Functions

Objective

Understanding Props, State, and Callback Functions in React

Tools Used

- VS Code

Link to Repository (Source Code): https://github.com/TALHA-089/WEB-LAB-07.git

Task 1: Teacher.jsx

Code

```jsx
import React from "react";

const Teacher = (props) => {
  return (
    <div style={{border:"1px solid blue", padding:"10px", margin:"10px"}}>
      <h2>Teacher: {props.name}</h2>
      <h3>Subject: {props.subject}</h3>
    </div>
  );
};

export default Teacher;
```

Figure 1. Code for Component Teacher.jsx

react-lab07 > src > App.jsx > ...

```jsx
import React from "react";
import Teacher from "./components/Teacher";

function App() {
  return (
    <div>
      <Teacher name="Engr. Subas Bilal" subject="Web Engineering" />
      <Teacher name="Engr. Salman Zafar" subject="Cloud Computing" />
    </div>
  );
}

export default App
```

Figure 2. Code for App.jsx

Screenshot

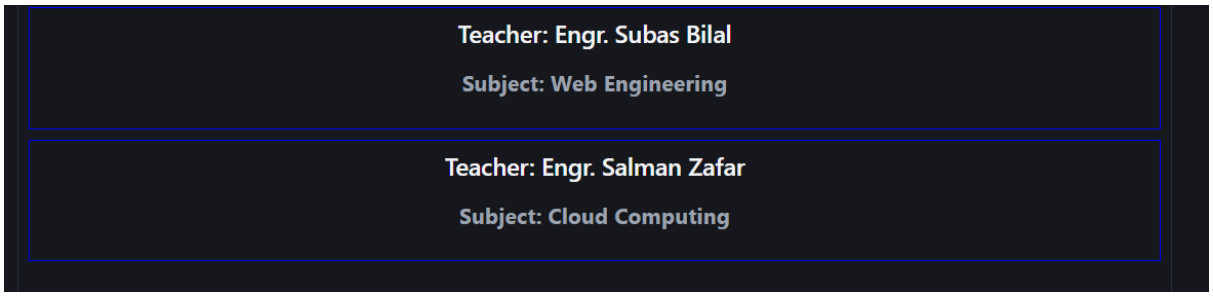

Figure 3. Output Screenshot

Task 2: ClickTracker.jsx

Code

```jsx
import React, { useState } from "react";

const ClickTracker = () => {

  const [clicks, setClicks] = useState(0);

  return (
    <div style={{margin:"20px"}}>
      <h2>Total Clicks: {clicks}</h2>

      <button onClick={() => setClicks(clicks + 1)}>
        Click Me
      </button>

    </div>
  );
};

export default ClickTracker;
```

Figure 4. Code for ClickTracker.jsx

react-lab07 > src > App.jsx > ...

```jsx
import React from "react";
import ClickTracker from "./components/ClickTracker";

function App() {
  return (
    <div>
      <ClickTracker />
    </div>
  );
}

export default App
```

Figure 5. Code for App.jsx

Screenshot

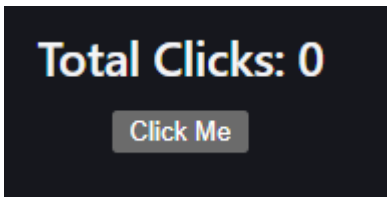

Figure 6. Initial State

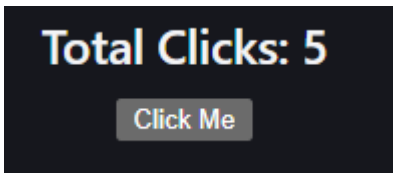

Figure 7. After 5 Clicks

Task 3: ResetCounter.jsx

Code

```react
import React from "react";

const ResetCounter = ({ onReset }) => {

  return (
    <button onClick={onReset}>
      Reset Counter
    </button>
  );

};

export default ResetCounter;
```

Figure 8. Code for ResetCounter.jsx

```react
import React, { useState } from "react";
import ClickTracker from "./components/ClickTracker";
import ResetCounter from "./components/ResetCounter";

function App() {
  const [count, setCount] = useState(0);

  const handleIncrement = () => setCount(count + 1);
  const handleReset = () => setCount(0);

  return (
    <div>
      <ClickTracker count={count} onIncrement={handleIncrement} />
      <ResetCounter onReset={handleReset} />
    </div>
  );
}

export default App;
```

Figure 9. Code for App.jsx

Screenshot

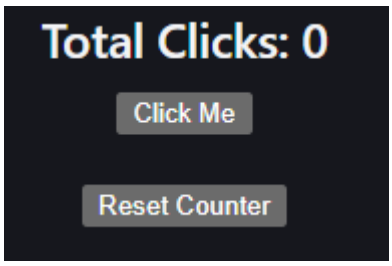

Figure 10. Initial State

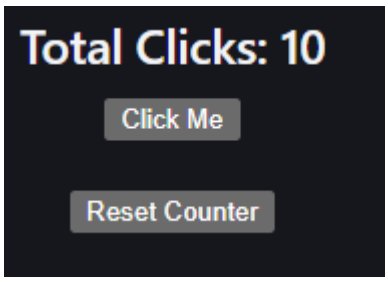

Figure 11. After 10 Clicks

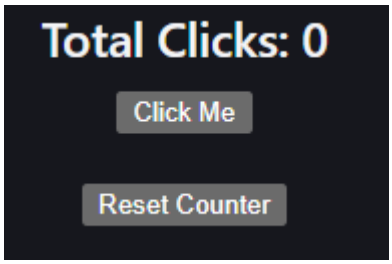

Figure 12. After Reset

Task 4: DecreaseCounter.jsx

Code

```jsx
import React from "react";

const DecreaseCounter = ({ onDecrease }) => {

  return (
    <button onClick={onDecrease}>
      Decrease Count
    </button>
  );

};

export default DecreaseCounter;
```

Figure 13. Code for DecreaseCounter.jsx

```jsx
import React, { useState } from "react";
import ClickTracker from "./components/ClickTracker";
import ResetCounter from "./components/ResetCounter";
import DecreaseCounter from "./components/DecreaseCounter";

function App() {
  const [count, setCount] = useState(0);

  const handleIncrement = () => setCount(count + 1);
  const handleReset = () => setCount(0);
  const handleDecrease = () => setCount(count - 1);

  return (
    <div>
      <ClickTracker count={count} onIncrement={handleIncrement} />
      <ResetCounter onReset={handleReset} />
      <DecreaseCounter count={count} onDecrease={handleDecrease} />
    </div>
  );
}

export default App;
```

Figure 14. Code for App.jsx

Screenshot

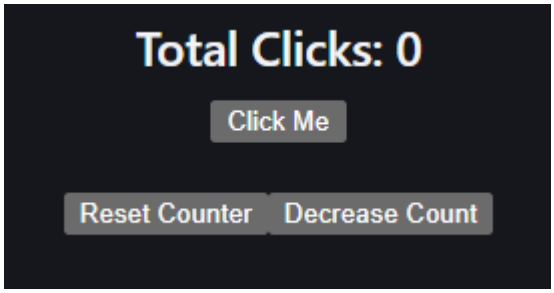

Figure 15. Initial State

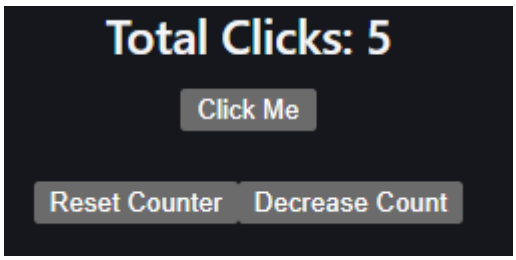

Figure 16. After 5 Clicks

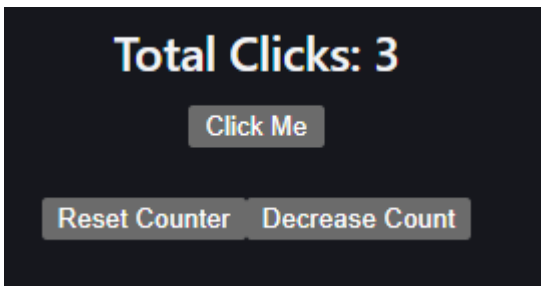

Figure 17. After Decreasing the Count by 2

Task 5: Child.jsx

Code

```jsx
import React from "react";

function Child({ onIncrement }) {

  return (
    <div>
      <button onClick={() => onIncrement(1)}>Increase by 1</button>
      <button onClick={() => onIncrement(5)}>Increase by 5</button>
    </div>
  );

}

export default Child;
```

Figure 18. Modified Code for Child.jsx

Screenshot

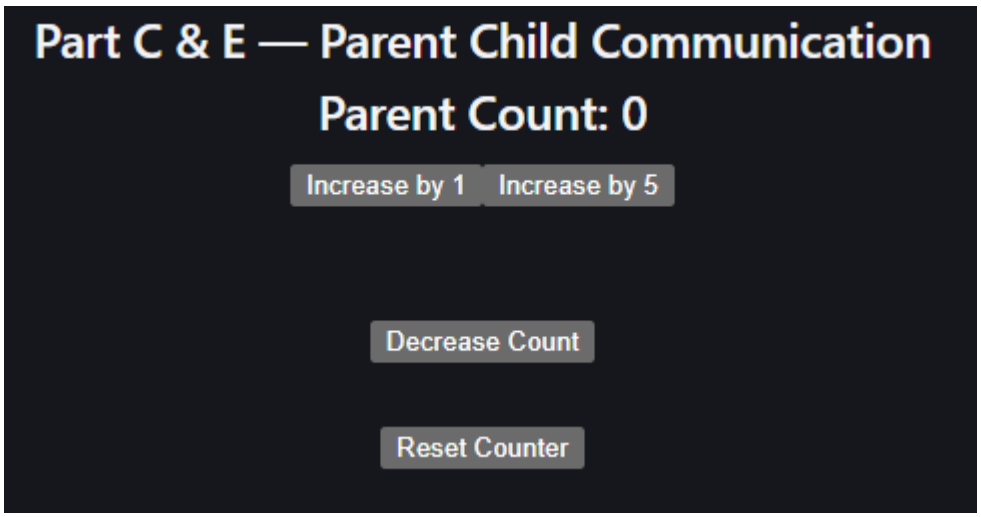

Figure 19. Initial State

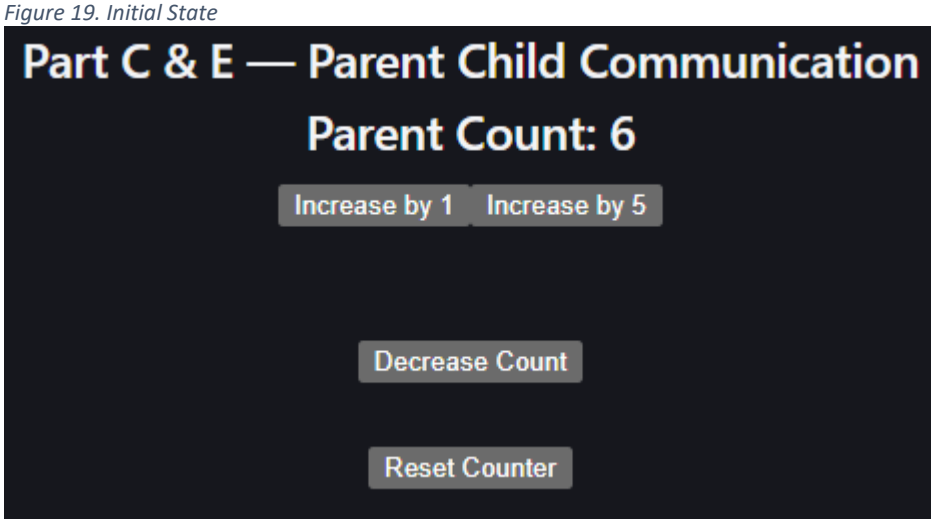

Figure 20. After Increase by 1 and Increase by 5

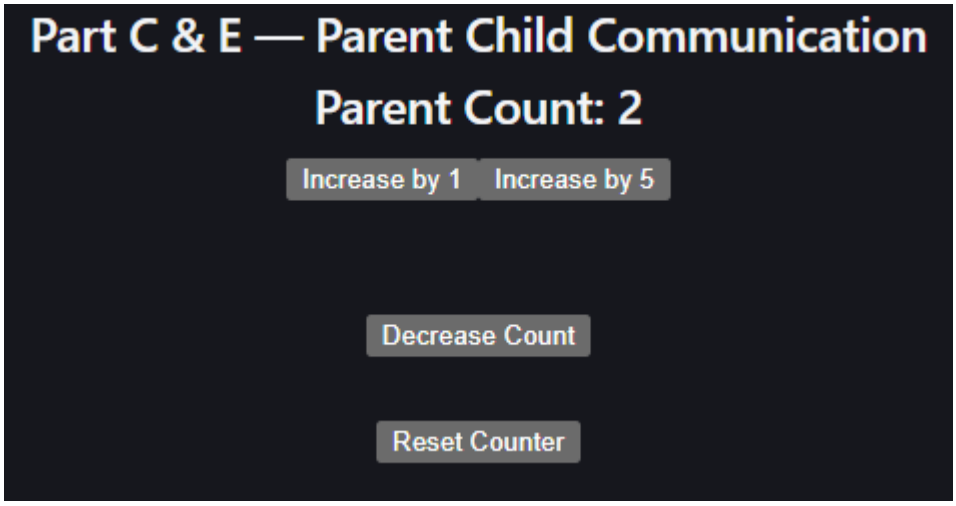

Figure 21. Decreasing the Counter by 4

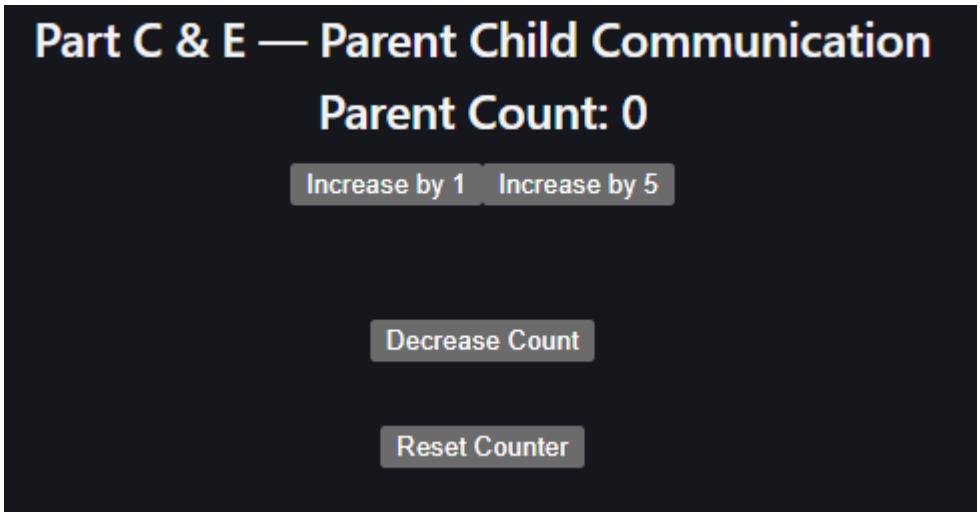

Figure 22. After Resetting